Khushi Kumari

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EDUCATION

Vellore Institute of Technology, Vellore May 2027
Bachelor of Technology in Information Technology **CGPA: 7.98/10**

WORK EXPERIENCE

AI/ML Intern | *National Informatics Centre (NIC), Patna* May 2026 – June 2026

- Developed a **Vision-Based Classroom Attendance System** using **YOLOv8** object detection on annotated educational datasets with 2 classes: students and teachers.
- Achieved **91.5% mAP50, 92.0% precision, and 86.3% recall**, automating classroom attendance monitoring through real-time object detection.
- Processed and annotated classroom image datasets using **Roboflow** and developed a production-ready **FastAPI**-based inference pipeline.
- Developed a **YOLOv8s-based water-body identification system** for village environments, classifying 4 categories: lakes, canals, handpumps, and wells.
- Internship Certificate

PROJECTS

Vision-Based Classroom Attendance System | *Python, PyTorch, YOLOv8, OpenCV, Flask, FastAPI* GitHub Link

- Built an AI-based student head detection and counting system using custom **YOLOv8**, achieving **92%+ mAP50 precision** and 30-50ms inference time.
- Engineered system to handle real-world challenges like crowding, blur, low light, and varied angles while excluding teachers to accurately count students.
- Developed end-to-end processing pipeline supporting batch image detection and Flask/FastAPI web interface.

Cognitive Sovereignty Enforcement System | *Android, Kotlin, Jetpack Compose* GitHub Link

- Built a notification control system using NotificationListenerService to regulate notifications across **15+ applications** through a token-based economy.
- Developed an **adaptive AI layer** analyzing 7-day behavioral patterns, dynamically adjusting app usage limits and reducing unnecessary notifications by **30%**.
- Automated daily token resets and restriction workflows using **WorkManager** for seamless background processing.

Judicial Outcome Prediction System | *PySpark, Hadoop, NLP, XGBoost* GitHub Link

- Built a Big Data pipeline using PySpark and HDFS to process **10,000+ legal records** for judicial outcome prediction.
- Applied NLP techniques including tokenization and Named Entity Recognition (NER) to extract features from large-scale legal datasets.
- Developed an XGBoost-based prediction model achieving **86% accuracy** with explainable AI techniques for model interpretability.

Fuzzy Concept-Bottleneck Skin Cancer Diagnosis | *Python, Flask, OpenCV, Fuzzy Logic, AWS* GitHub Link

- Developed an interpretable skin-lesion diagnosis pipeline combining segmentation, geometric features, fuzzy concepts, and concept-bottleneck modeling for explainable diagnosis.
- Prepared **2,594 ISIC 2018 image-mask pairs** (1,815 training, 389 validation, and 390 test samples) for lesion segmentation.
- Built a Flask web application with interactive ABCDE fuzzy concepts and verified deployment across **10 AWS services** including EC2, S3, RDS, IAM, CloudWatch, and API Gateway.

SKILLS

Programming Languages:	Python, Java, C++, JavaScript, SQL, Kotlin
ML & AI:	YOLO (YOLOv5, YOLOv8), OpenCV, scikit-learn, XGBoost, NLP, NER, PySpark, Hadoop, Pandas, NumPy
Web Technologies:	Node.js, Express.js, REST APIs, HTML, CSS
Databases:	MongoDB, SQL (MySQL), DynamoDB
Cloud Platforms:	AWS (EC2, S3, RDS, IAM, CloudWatch, API Gateway, Secrets Manager, Cognito, SNS)
Tools & Frameworks:	Git, GitHub, VS Code, Flask, FastAPI

ACHIEVEMENTS

- Secured **1st Place** in Script Showdown, Yantra 2026 (VIT) — Certificate

- Awarded **Certificate of Appreciation** for contribution to RC Cars Building & Racing Event (SME, VIT).
- Successfully completed **Full Stack Development Virtual Internship** at Cognifyz Technologies.

ADDITIONAL INFORMATION

- **Relevant Coursework:** Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Cloud Computing, Machine Learning
- **Areas of Interest:** Machine Learning, Computer Vision, Artificial Intelligence, Backend Development
- **Extracurricular Activities:** IET-VIT & Creativity Club Member (2023 – 2024)